

18 – Core Data Model

Logical data model for the platform: modelling principles; entity, location and tenancy scoping; catalogue, batch and price ceilings; the immutable document base type with numbering series and compensating corrections; the append-only audit log; the event backbone and sync state; and the module schema register with the cross-module invariants the module designs introduced.

Field	Value
Document	18 – Core Data Model (design artefact; Stage 1)
Version	v0.3.1 – consolidated: incorporates v0.2 and the Data Model Consolidation Addendum (18B) in one document; 18B is withdrawn. v0.3.1 restores v0.2 material dropped in the v0.3 rewrite (visibility guarantees, audit notes, decision consequences, full abbreviation list)
Date	18 September 2026
Status	Design of record. Parts A–E are authoritative for kernel and shared tables; Part F registers every module schema and names the implementation guide whose DDL is authoritative for module-owned tables.
Inputs	02 Domain Model, 03 Document State Machines, 05 SRS, 06 NFR, 07 Matrices, 09 Decisions Log, 16 Questionnaire Answers, 17 Solution Architecture v0.2.1, 19 Kernel Services v0.2, 32 Sync Contract, 24B Numbering Addendum, the module designs and guides 21–29 and 21A–29A, the September 2026 review decisions and the multi-MRP study.
Audience	Engineering leads and database designers. Business readers need only Sections 1, 2 and Part F.
Scope boundary	Logical model at table and column level, with invariants. Physical tuning belongs to the database design and doc 35.

How to read this document. Every table lists its columns, types and the rules that bind them. Rules spanning tables are numbered invariants at the end of each part; they are the contract application code, database constraints and tests must all honour. Where a module design changed a rule from v0.2, the change is stated in place. Abbreviations are expanded on first use and collected in Section 10.

1. Modelling principles

Ten rules govern every table in this model. Where a table departs from one, the departure is stated.

#	Rule	Statement
P1	One owning entity per row	Every operational row carries <code>owner_entity_id</code> . A document between two legal entities carries both parties, but exactly one of them is the owner (the issuer). Tenancy scoping, visibility and audit all key off this column.
P2	Identifiers are generated where the record is born	Primary keys are UUID version 7: globally unique, time-ordered, safe to generate on a disconnected till. Business-facing numbers are separate attributes.
P3	Documents are immutable once issued	After issuance a document header and its lines are never updated or deleted. Any change is a new document that references the original (Part C).
P4	Ledgers are append-only	Stock movements, account postings, document state history, the outbox and the audit log are insert-only. Balances are derived, never edited. The application role holds INSERT only.
P5	Reference data is effective-dated	Prices, tax rates, control prices, relationships and conversions carry <code>effective_from</code> / <code>effective_to</code> ; a past transaction can always be shown to have been correct at the time.
P6	One writer per record type	The till is the only writer of its own sale, session, receiving and count records; central is the only writer of everything else, pushed to tills read-only. No merge logic exists (ADR-10).
P7	Trilingual text with English as the anchor	<code>name_en</code> mandatory; <code>name_si</code> , <code>name_ta</code> optional except where a status requires them; ICU collations on each.
P8	Money and quantity are exact	Money <code>numeric(14, 2)</code> LKR; unit prices and costs <code>numeric(14, 4)</code> ; quantities <code>numeric(14, 3)</code> in the base unit. No floating point.
P9	Time is stored twice	<code>occurred_at</code> in UTC and <code>occurred_local</code> as the device reading; <code>received_at</code> stamped by the server; sequence numbers, not clocks, order events.
P10	Nothing is hard-deleted	Reference rows carry a status; users are deactivated; documents are voided by compensating documents. The only DELETE privilege belongs to migration tooling.

2. Model overview

Part	Owning module (doc 17 v0.2.1)	Tables	Depends on
A – Entity, location, tenancy	M1 Party, Tenancy & Security	<code>entity</code> , <code>entity_relationship</code> , <code>location</code> , <code>till_position</code> , <code>device</code> , <code>app_user</code> , <code>permission</code> , <code>role</code> , <code>role_permission</code> , <code>user_role</code> , <code>sod_pair</code> , <code>external_grant</code>	–
B – Catalogue, batch, ceilings	M2 (catalogue, batch, images); M3 (price lists, rules, control prices, policies); M5 (lots, movements, recipes)	<code>uom</code> , <code>supplier</code> , <code>sku</code> , <code>sku_uom_conversion</code> , <code>sku_barcode</code> , <code>sku_image</code> , <code>sku_alias</code> , <code>tag</code> , <code>sku_tag</code> , <code>tax_category</code> , <code>tax_rate</code> , <code>batch</code> , <code>location_assortment</code> , <code>stock_lot</code> , <code>stock_movement</code> , <code>control_price</code> , <code>price_list</code> , <code>price_list_line</code> , <code>discount_rule</code> , <code>mrp_policy</code> , <code>repack_recipe</code>	A
C – Document base type	Kernel (document types owned by M4, M5, M6, M7)	<code>document_type</code> , <code>numbering_series</code> , <code>document</code> , <code>document_line</code> , <code>document_link</code> , <code>document_state_history</code> , <code>document_attachment</code>	A, B
D – Audit log	Kernel	<code>audit_event</code> , <code>audit_event_type</code>	A
E – Event backbone and sync state	Kernel	<code>event_outbox</code> , <code>event_inbox</code> , <code>device_sync_cursor</code> , <code>sync_quarantine</code> , <code>snapshot_change_log</code> , <code>idempotency_key</code>	A
F – Module schema register	All modules	Every module-owned table by schema, with its authoritative DDL; cross-module invariants; document type ownership	A–E

Module-specific tables (M4 extensions, M5 tasks and thresholds, M7 customers and accounts, M8 projections, M9 clients and exports) are introduced by their module designs and registered in Part F; the module implementation guide's section 3 is the authoritative DDL for them. A module design raises a change request here only when a shared table changes. The till carries local copies of parts C–E following the same logical model (Section 7).

3. Part A – Entity, location and tenancy scoping

Three kinds of legal entity and one kind of place. A **shop is a location, not an entity**: it holds stock and issues receipts, but the legal party to every transaction is the MPCs or distributor that owns it. This distinction is enforced by the absence of legal attributes on location.

Table entity

A legal party: the Federation, a distributor (a large MPCs with a distribution role) or an MPCs. About 876 rows at go-live.

Column	Type	Meaning and rules
entity_id	uuid PK	UUIDv7.
entity_code	varchar(12) UNIQUE	Human-facing prefix in document numbers, e.g. M042; immutable.
entity_type	enum	FEDERATION DISTRIBUTOR MPCs; exactly one FEDERATION row.
legal_name_en / _si / _ta	text	Registered name; English mandatory (P7).
registration_no	varchar(40)	Co-operative registration number.
vat_registration_no	varchar(20)	Required for all chain entities (J-03); printed on tax invoices of both parties.
district	varchar(40)	One of the 25 districts.
financial_year_start_month	smallint	Common federation-wide (J-08) but stored per entity.
default_language	enum	en si ta.
costing_policy	enum	WEIGHTED_AVERAGE only in phase 1 (E-03).
credit_exposure_warn_only	boolean	Always true in phase 1 (ADR-12); policy as data.
responsible_officer_user_id	uuid FK app_user (nullable)	The entity's named responsible officer under the PDPA and the data-governance agreement (FR-GOV-060); required for activation (CR-21-1).
data_governance_signed_on	date (nullable)	Date the entity signed the data-governance agreement (M-11) (CR-21-1).
status	enum	ONBOARDING ACTIVE SUSPENDED; no CLOSED state (M-07).

- The Federation row is the only entity whose users may hold the federation-wide read-only view (3.7).
- There is no parent/child column between entities; trading tiers are expressed by entity_relationship.

Table entity_relationship

One row per trading pair; the home of bilateral commercial terms (A-06, A-07, A-09, D-06).

Column	Type	Meaning and rules
relationship_id	uuid PK	
seller_entity_id / buyer_entity_id	uuid FK entity	seller <> buyer enforced.
price_list_id	uuid FK price_list	The effective-dated TRADE list (Part B); must belong to the seller.
credit_limit	numeric(14,2)	Informative (ADR-12); changes permission-gated with MFA and audited.
payment_terms_days	smallint	Ageing basis.
discrepancy_window_days	smallint	D-06 time limit.
order_lock_hours_before_eta	smallint	C-08 lock window; per-order override on the order.
allocation_rule	enum	FCFS default; per seller; audited on change (C-03, C-04).
status	enum	DRAFT ACTIVE SUSPENDED.
effective_from / effective_to	date	Effective-dated: an amendment closes a row and opens another (P5).

- A-I3: at most one ACTIVE row per (seller, buyer) at any date (exclusion constraint). Intra-entity movements are never relationships.

Table location

A physical place belonging to exactly one entity: warehouse, shop or office. A shop is one stock location (E-02).

Column	Type	Meaning and rules
location_id	uuid PK	
owner_entity_id	uuid FK entity	Immutable (A-I2).
location_code	varchar(12)	Unique within the owner; used in location-scoped series, e.g. M042-S07.
location_type	enum	WAREHOUSE SHOP OFFICE; only the first two hold stock.
name_en / _si / _ta	text	
address, district, geo_lat, geo_lng	text / numeric	Drops and rollout planning.
language	enum	Per-shop interface and receipt language (P-02, H-09); null inherits the entity default.
trading_hours	jsonb	Update windows (doc 31) and day-close.
size_band	enum	S M L (H-06).
connectivity_spec_met	boolean	Go-live gate (Q-06); required for ACTIVE.
primary_till_position_id	uuid FK till_position	Holds the location-scoped numbering series; reassignment audited (A-I7).
status	enum	PLANNED ONBOARDING ACTIVE DORMANT.

Table till_position

A logical checkout lane; till-issued document series bind to the position, not the hardware (ADR-11).

Column	Type	Meaning and rules
till_position_id	uuid PK	
location_id	uuid FK location	Must be a SHOP.
position_no	smallint	Unique per location; appears in receipt numbers.
status	enum	ACTIVE RETIRED; retired positions keep their series; numbers never reused.

Table device

A physical terminal or workstation; assignment to a position is the only link between hardware and numbering.

Column	Type	Meaning and rules
device_id	uuid PK	
hardware_serial	varchar(64) UNIQUE	Enrolment identity.
device_kind	enum	POS_TERMINAL WORKSTATION DRIVER_MOBILE.
owner_entity_id	uuid FK entity	P1.
current_till_position_id	uuid FK till_position	At most one device per position (partial unique index); history in the audit log.
enrolled_at, last_seen_at	timestampz	Fleet health; freshness.
app_version	varchar(20)	Floor enforcement (ADR-24).
status	enum	ENROLLED ACTIVE SUSPENDED RETIRED.

Users, roles, permissions, segregation

Table	Key columns	Rules
app_user	user_id PK; home_entity_id; username UNIQUE; display_name; language; user_kind BACK_OFFICE TILL BOTH EXTERNAL; provider_subject; pin_hash; pin_history; sandbox; succeeds_user_id; status	One home entity. Till operators authenticate by PIN (Argon2id hash distributed in the operator snapshot); back office by the identity provider with TOTP for financial and administrative functions. Deactivated, never deleted; a returning member of staff gets a new record linked by succeeds_user_id.

Table	Key columns	Rules
permission	permission_code PK; module; description_en; offline_allowed; requires_mfa; scope LOCATION ENTITY FEDERATION; limits_schema	The catalogue, versioned as a whole (permission_catalogue_version.rv). offline_allowed is what the till caches; requires_mfa marks online-only actions; scope is the finest grantable level.
role	role_id PK; owner_entity_id (null = federation template); names; is_template; role_class OWN FEDERATION_VIEW EXTERNAL_TIMEBOXED; template_role_id; template_version_seen; version; status	Templates clone on adoption; drift is notified, not forced (doc 19 DR-4).
role_permission	role_id; permission_code; limits jsonb	Limits carry approval ceilings validated against the permission's limits_schema.
user_role	user_id; role_id; scope_entity_id; scope_location_id (nullable)	Scope columns implement M-08: an MPCS administrator grants roles scoped to its shops; the Federation holds no downward administrative scope.
sod_pair	sod_pair_id PK; permission_a; permission_b; mode INSTANCE ROLE; owner_entity_id (null = federation default)	CR-21-2. INSTANCE: the same user may hold both but not act as both on one document; ROLE: may not coexist in one role. Federation defaults: writeoff.request/approve, writeoff.request/witness, adjust.request/approve, user.manage/audit.review-self, cus.account.adjust/adjust_approve.
external_grant	grant_id PK; grantee_user_id; scope_entity_ids[]; valid_from; valid_until (≤ 12 months); reason; status	Regulator and auditor access (M-10, J-09); every grant and login audited.

3.7 Tenancy scoping and visibility enforcement

One shared database; ownership is a property of rows. Every database session carries the caller's scope as settings (app.scope_entity_id, app.scope_location_id, app.scope_class, app.granted_entities, set with SET LOCAL per transaction); row-level security policies read them. A transaction without a scope fails closed.

Class	Who	Reads	Writes
OWN	Any user in scope	Rows where owner = scope entity (and location when the assignment is location-scoped)	Per permission
PARTY	Seller or buyer of a document	Trading-facing columns of documents where the scope entity is a party, through masking views that omit cost, margin and internal notes	Their own side's commands
FEDERATION_VIEW	Federation roles marked view-all	Everything, including margins and shrinkage (M-02 to M-04)	None
EXTERNAL_TIMEBOXED	Regulator, auditors	Granted scope, read-only, until valid_until	None
API_CLIENT	OAuth2 clients (M9)	Mapped to OWN or PARTY for the client's entity within its scopes	One phase-1 write (journal acknowledgement)

What the classes guarantee: no query runs without a declared scope (a missing scope fails closed); a distributor cannot see an MPCS's internal data beyond the documents they trade (FR-GOV-020); a shop sees nothing of a sibling shop (M-05) because shop-scoped users carry a location scope and shop-internal tables filter on it; the buyer sees the seller's invoice, never the seller's cost basis, because PARTY reads go through masking views; the Federation and external grantees read everything in their scope and write nothing.

Invariants for Part A.

- A-I1. Every operational table has a non-null owner_entity_id or derives one through a non-null location_id.
- A-I2. location.owner_entity_id is immutable after creation.
- A-I3. At most one ACTIVE entity_relationship per (seller, buyer) at any date.
- A-I4. A till_position belongs to exactly one SHOP and its position_no is unique within it; positions are retired, never deleted.
- A-I5. A device holds at most one till position at a time; the change is an audited event.

- A-I6. No user holds a role scoped outside their home entity unless the role class is `FEDERATION_VIEW` or `EXTERNAL_TIMEBOXED`.
- A-I7. A SHOP has a `primary_till_position_id` before it can issue any location-scoped document; only the primary till issues them (ADR-47).
- A-I8. No grantor confers a permission they do not hold; an entity never loses its last holder of `gov.user.manage`; an API client's scopes never exceed its grantor's permissions (I-F9).

4. Part B – Catalogue, units, batches and price ceilings

Three ideas: the **SKU** is what is counted and sold; the **batch** is a physical supply identity carrying expiry and the printed MRP; the **stock lot** is an entity's holding of a batch at a location, in a condition, carrying that entity's cost. Separating batch from lot lets one physical batch be owned successively by the Federation, a distributor and an MPCS at three different costs (ADR-38).

Table uom

Federation-governed units of measure.

Column	Type	Meaning and rules
uom_code	varchar(10) PK	EA, KG, G, L, ML, CASE, PKT, BAG, BOX, DOZ.
name_en / _si / _ta	text	
is_weight	boolean	Drives the weigh-station path (B-06, B-11).

Table supplier

Minimal supplier identity until M10 Procurement absorbs it in phase 2 (CR-22-1).

Column	Type	Meaning and rules
supplier_id	uuid PK	
owner_entity_id	uuid FK entity	The registering entity.
name	text	Unique per owner.
status	enum	ACTIVE INACTIVE.

Table sku

The smallest distinctly tracked and counted item, in one global namespace (B-01).

Column	Type	Meaning and rules
sku_id	uuid PK	
sku_code	varchar(12) UNIQUE	System-generated, opaque, never reused; used for till code lookup (B-11).
owner_entity_id	uuid FK entity	Federation for SHARED; the creating entity for DRAFT and LOCAL (B-08).
status	enum	DRAFT LOCAL SHARED INACTIVE; prior_status kept for reactivation. INACTIVE keeps selling lots on hand when sell_through applies (doc 22 DR-3).
short_name_en	varchar(80) NOT NULL	Mandatory (P7); _si, _ta mandatory for SHARED.
description_en / _si / _ta	text	
base_uom_code	FK uom	KG when sold_by_weight.
sold_by_weight, batch_tracked, expiry_tracked	boolean	Expiry tracking implies batch tracking. Non-batch-tracked SKUs get a synthetic batch per receipt line so MRP and cost always have a home (ADR-38).
has_printed_mrp	boolean	New in v0.3 (doc 22 DR-7): false for loose produce and unpriced goods, making a null batch MRP legitimate rather than missing.
expiry_warning_days	smallint	Category default copied at creation.
tax_category_id	uuid FK tax_category	Per-SKU tax category (G-09).
multi_mrp_policy	enum	AUTO_LOWEST BARCODE_RESOLVED PICKER; denormalised effective value maintained from M3's mrp_policy (ADR-41).
origin_kind	enum	PURCHASED REPACK_OUTPUT LOCAL_SOURCED.
attributes	jsonb (≤ 4 KB)	Bounded, entity-owned free attributes; non-searchable; no behaviour (doc 20 Appendix A).

Table sku_uom_conversion

Pack conversions for the same physical item (B-02, ADR-07); a genuinely different retail item is a second SKU via a repack recipe.

Column	Type	Meaning and rules
sku_id, uom_code	PK	
factor_to_base	numeric(14,6)	One unit = factor × base units.
effective_from, effective_to	date	Exclusion constraint per (sku, uom) (P5).

Table sku_barcode

Many barcodes to one SKU (B-12); the optional batch link makes barcode-resolved MRP possible.

Column	Type	Meaning and rules
barcode, symbology	varchar(48), enum	EAN13 EAN8 UPCA GS1_128 GS1_DATAMATRIX GS1_QR INTERNAL. Factory symbologies unique federation-wide; INTERNAL unique per owner (B-12).
sku_id, uom_code	FK	A case barcode maps to the CASE unit of the same SKU.
batch_id	uuid FK batch (nullable)	Set when the code identifies one batch: batch sticker, MRP-wise manufacturer code, or a 2D code carrying application identifier 10.
owner_entity_id	uuid FK entity	In-store codes are owned by the registering entity.
status	enum	ACTIVE RETIRED.

- In-store 20–29 labels encode SKU code plus weight or batch, never a price, so the shared engine and the control-price ceiling always apply (ADR-43).

Table sku_image

Product images (September 2026 decision): one per SKU, optionally one per barcode; tills cache thumbnails only.

Column	Type	Meaning and rules
image_id	uuid PK	
sku_id, barcode (nullable)	FK, varchar(48)	Per pack artwork when set.
owner_entity_id	uuid FK entity	Federation for SHARED; an entity may hold a local override for its own locations.
object_key_full, object_key_thumb	text	Thumbnail about 128 px, under 10 KB, rides in the shop snapshot.
content_hash	char(64)	
status	enum	PENDING ACTIVE RETIRED; one ACTIVE per (sku, barcode-or-null, owner).

Table sku_alias

Preserves an originating entity's history when a local SKU is promoted or merged (B-10, FR-SKU-100).

Column	Type	Meaning and rules
alias_sku_id	uuid PK FK sku	The retired or promoted local SKU.
canonical_sku_id	uuid FK sku	Reports resolve aliases at query time.
effective_from	date	

Tags, tax, assortment

Table	Key columns	Rules
tag`, `sku_tag	tag_code PK; names; governed; owner_entity_id (null for governed)	A governed vocabulary replaces a mandatory hierarchy (B-15). Policies in M3 and M5 (expiry windows, tolerances, thresholds, control-price scope, multi-MRP policy) attach to governed tags only; local tags carry no behaviour. The governed tag core-range marks goods every shop can sell.

Table	Key columns	Rules
tax_category`, tax_rate	tax_category_id PK; code; rate_percent; effective_from/to	Standard VAT only (J-03); exclusion constraint per category; publication carries apply_from to tills.
location_assortment	location_id; sku_id; reasons[]; sell_through; since	CR-22-2. The set of SKUs a location's till must know: retail-list SKUs $\cup$ SKUs with a lot at the location $\cup$ the owner's LOCAL SKUs $\cup$ core-range (I-F1). Materialised by M2 consumers; reconciled nightly.

Table batch

A physical supply identity for a SKU, created when stock first enters the federation. Global, not owned by the holder.

Column	Type	Meaning and rules
batch_id	uuid PK (with created_at partition key)	
sku_id	uuid FK sku	
batch_no	varchar(40)	As printed; unique with (sku, supplier). A near-duplicate number for the same SKU and supplier is a steward suspect, not an error.
supplier_id	uuid FK supplier	Batch-level supplier attribution (B-03).
manufacture_date, expiry_date	date	Expiry mandatory when the SKU is expiry-tracked.
printed_mrp	numeric(14,2)	The MRP printed on the pack (ADR-08). Non-null when the SKU has has_printed_mrp and the receipt line quantity is positive; null otherwise is legitimate (I-F2). Immutable: a mis-keyed MRP is corrected by a replacement batch.
origin_document_id	uuid FK document	The GRN or repack that created it.
is_synthetic	boolean	True for non-batch-tracked SKUs (one per receipt line).
corrects_batch_id	uuid (nullable)	Set on a replacement batch; the original becomes SUPERSEDED and lots re-point (B-I8).
status	enum	REGISTERED SUPERSEDED.
created_at	timestamp tz	Partition key (monthly).

Table stock_lot

An entity's holding of one batch at one location in one condition: the unit of costing and of on-hand balance (M5).

Column	Type	Meaning and rules
stock_lot_id	uuid PK	
location_id	uuid FK location	Owner entity derived (A-I1).
batch_id	uuid FK batch	
condition	enum	GOOD DAMAGED (new in v0.3, doc 25 DR-2): damaged units received on a GRN form a DAMAGED lot at the same unit cost, sellable only through a markdown rule or written off. Unique per (location, batch, condition) (I-F3).
qty_on_hand	numeric(14,3)	In base units. May be negative (E-09, ADR-45): every negative occurrence is an exception item; negative_since and negative_acknowledged_at track it.
unit_cost	numeric(14,4)	This entity's acquisition cost: trade price at GRN (ADR-03), keyed cost for local supply, recipe-derived for repack outputs, restated by landed cost in phase 2.
received_at	timestamp tz	Ageing (E-11).
last_movement_seq	bigint	Idempotent re-application on replay.

- Weighted-average cost per (entity, SKU) is a separate M5 table (entity_sku_cost); the lot keeps its acquisition cost, the entity average is the costing figure (E-03, IAS 2).

Table stock_movement

Append-only ledger of every quantity change; the only way stock changes (M5).

Column	Type	Meaning and rules
movement_id	uuid PK (with received_at)	
source, movement_seq	text, bigint	Dense per (location, source) where source is a device id or 'central' (I-F4); the tie-breaker for ordering offline events.
location_id, batch_id, condition	FK, enum	
qty_delta	numeric(14,3)	Signed, base units.
movement_type	enum	RECEIPT SALE SALE_REVERSAL TRANSFER_OUT TRANSFER_IN REPACK_CONSUME REPACK_PRODUCE COUNT_ADJUST WRITE_OFF RETURN_TO_SELLER OPENING_BALANCE GRN_REVERSAL. Loss categories are on the write-off document.
unit_cost_at_movement	numeric(14,4)	Cost basis frozen at the time (entity average for issues).
document_id, document_line_id	uuid FK	Every movement cites its document (B-I5).
occurred_at, occurred_local, received_at	timestamp tz	P9.
operator_user_id, device_id	uuid FK	Attribution (H-13).

Table control_price

Government-gazetted maximum price: a date-effective, SKU- or tag-scoped ceiling that binds all stock regardless of the MRP printed on older packs (CAA Act s.20(5)); M3-owned, Federation-written (ADR-41).

Column	Type	Meaning and rules
control_price_id	uuid PK	
sku_id tag_code	uuid varchar(40)	Exactly one set; tag scope covers gazettes that name a category; where both a SKU and a tag ceiling are in force the lower applies (doc 23 DR-2, pending Federation legal).
ceiling_price, ceiling_uom_code	numeric(14,2), FK uom	Tax inclusive; converted through conversions.
effective_from, effective_to	date	Publication date; null while in force; exclusion constraint per scope (B-16).
gazette_reference	varchar(80)	Receipt footer and audit.
entered_by, entered_at		Federation catalogue steward only; every insert audited; never updated except effective_to.

- Enforcement is a hard block at authoring, at label printing and at sale (B-I7); tills cache the table and warn when the snapshot is stale.

Price lists, rules, policies, recipes (M3 and M5; summarised because Part C cites them)

Table	Key columns	Rules
price_list	price_list_id PK; owner_entity_id; kind TRADE RETAIL ADVISORY; name; version; source_version_id; status DRAFT PUBLISHED SUPERSEDED WITHDRAWN; apply_from	TRADE lists price a relationship; one PUBLISHED RETAIL list per MPSCS for all its shops (G-01); ADVISORY lists never resolve a price.
price_list_line	price_list_id; sku_id; uom_code; tier_from_qty; price; effective_from/to	Effective-dated with an exclusion constraint per (list, sku, uom, tier). Authoring rejects a price above any in-stock batch MRP at the owner's locations or above a control price in force (B-I7).
discount_rule	rule_id PK; owner_entity_id; name; kind TIME_LIMITED_PRICE QUANTITY_BREAK BILL_THRESHOLD FREE_ITEM EXPIRY_MARKDOWN; predicate jsonb; benefit jsonb; priority; advisory; valid_from/to; status	The sole mechanism for any reduction at a till (ADR-14); a closed vocabulary validated by JSON schema per kind; every reduction stamps its rule id so markdown cost is measured separately (F-07).
mrp_policy	policy_id PK; sku_id tag_code; policy; picker gap amount / percent; owner_entity_id	SKU scope beats tag; default AUTO_LOWEST (ADR-41).

Table	Key columns	Rules
adjustment_type	code PK; kind SPECIAL_DISCOUNT DELIVERY_FEE; permission_code; bounds	Manual adjustments exist only on Federation→distributor orders against a registered type (A-06).
repack_recipe	recipe_id PK; owner_entity_id; input_sku, input_qty; output_sku, output_qty; expected_loss_pct; status	Bulk-to-pack transformation (B-06, ADR-07); variance measured against expectation, not booked as shrinkage (M5).

4.9 Price resolution at the till (data-level statement)

For a scanned line the effective selling price is **min(list price, batch term, control price in force)** where the batch term is the lowest printed MRP among batches with positive on-hand at the location (AUTO_LOWEST), the scanned batch's MRP (BARCODE_RESOLVED) or the operator's pick (PICKER, only when the candidate gap exceeds the threshold), and is absent for SKUs without a printed MRP; then discount rules apply (one line rule by priority, one bill rule); then the control price is re-checked as a hard cap; then tax is decomposed and cash tenders round to the rupee at receipt level. Every receipt line stores `batch_id`, `mrp_applied`, `control_price_applied`, `cap_reason`, `rule_id` so its legality is reproducible from the row alone. The same library computes it at central and on the till (ADR-31).

Invariants for Part B.

- B-I1. `sku.short_name_en` is non-null for every status; `_si` and `_ta` are non-null for SHARED.
- B-I2. A factory barcode is unique per symbology federation-wide; INTERNAL codes are unique within their owner.
- B-I3. Every `stock_lot` references a batch of the same SKU as the receiving document line; unique per (location, batch, condition).
- B-I4. `stock_lot.qty_on_hand` equals the sum of movements for that lot; a nightly reconciliation asserts it and raises an exception on drift, never fixes.
- B-I5. No `stock_movement` exists without a `document_id`.
- B-I6. At most one `control_price` in force per scope per date.
- B-I7. A price-list line, rule benefit or printed label that would exceed a control price in force is rejected at authoring or printing, not merely at sale.
- B-I8. A batch's `printed_mrp` is immutable; corrections create a replacement batch and re-point lots.
- B-I9. A batch's `printed_mrp` is non-null exactly when the SKU has `has_printed_mrp` and the line quantity is positive (I-F2).

5. Part C – The document base type

Every trading and stock artefact is a document: order, delivery note, GRN, discrepancy, claim, invoice, credit and debit note, payment receipt, customer payment receipt, till receipt, write-off, count, adjustment, repack, transfer, opening balance. They share one header table, one line table and one lifecycle discipline; type-specific data lives in extension tables in the owning module's schema, keyed one-to-one on `document_id`, inheriting immutability and partitioning (I-F10).

Table `document_type`

Registry of document kinds and their numbering behaviour; flags as decided in 24B.

Column	Type	Meaning and rules
<code>doc_type_code</code>	<code>varchar(8)</code> PK	ORD, DN, GRN, DISC, CLM, INV, CN, DN2, PRC, CPR, RCT, WOF, CNT, ADJ, RPK, XFR, OPB.
<code>name_en / _si / _ta</code>	<code>text</code>	Printed heading.
<code>series_scope</code>	<code>enum</code>	ENTITY LOCATION TILL_POSITION (5.5).
<code>issuer_role</code>	<code>enum</code>	BUYER SELLER HOLDER.
<code>bilateral</code>	<code>boolean</code>	Counterparty mandatory for ORD, DN, GRN (when against a DN), DISC, CLM, INV, CN, DN2, PRC.
<code>fiscal</code>	<code>boolean</code>	True for RCT, INV, CN, DN2, PRC, CPR (24B); the others are protected by the same mechanism but their gaps are operational findings.
<code>offline_issuable</code>	<code>boolean</code>	RCT, GRN, CNT, RPK, XFR-receipt, CPR and WOF-draft are offline-issuable on the primary till (LOCATION series) or any till (TILL_POSITION series); ORD, INV, CN, DN2, PRC are not (H-02/H-03).
<code>posting_map</code>	<code>jsonb</code>	Journal-posting roles per line kind and side (J-02); registered into M9's <code>posting_map_registry</code> .
<code>owning_module</code>	<code>text</code>	M4, M5, M6 or M7 (Part F).

Table `numbering_series`

One row per series; sequential and gapless within the series; numbers final at issuance and never reused; voids keep their numbers (ADR-11, 24B N-3, N-4).

Column	Type	Meaning and rules
<code>series_id</code>	<code>uuid</code> PK	
<code>doc_type_code</code>	FK <code>document_type</code>	
<code>owner_entity_id</code>	<code>uuid</code> FK entity	Always set.
<code>location_id, till_position_id</code>	<code>uuid</code> FK (nullable)	Set per scope.
<code>prefix</code>	<code>varchar(24)</code>	{entity_code}-{location_code}-{T{position}}-{type}; display {prefix}-{number:07d}; no year segment (24B N-1).
<code>next_number</code>	<code>bigint</code>	Central view; for offline scopes the last synchronised value – the holder device holds the authoritative counter.
<code>holder_device_id</code>	<code>uuid</code> FK device (nullable)	LOCATION scope: the shop's primary till; TILL_POSITION scope: the assigned device; null for ENTITY.
<code>fy_reset</code>	<code>boolean</code>	False everywhere (24B N-1).
<code>status</code>	<code>enum</code>	ACTIVE CLOSED; retiring a position closes its series.

- Pre-reserved number ranges were evaluated and rejected (ADR-11). Each series has one counter with one writer.

Table `document`

The shared header; immutable after `issued_at` (P3).

Column	Type	Meaning and rules
<code>document_id</code>	<code>uuid</code> PK	UUIDv7 generated by the issuing device; present from draft.

Column	Type	Meaning and rules
doc_type_code, series_id, doc_number, doc_number_display		Series and number null while draft; display frozen at issuance.
owner_entity_id, counterparty_entity_id	uuid FK	Issuer; counterparty mandatory when bilateral.
location_id, till_position_id, device_id	uuid FK (nullable)	
status	enum	Type-specific (doc 03; docs 24–27); the base enforces DRAFT before issuance and terminal states after.
issued_at, issued_local, business_date	timestamptz, date	Immutability begins at issued_at; business_date from the location's day-close, never the clock.
operator_user_id	uuid FK app_user	H-13.
currency	char(3)	LKR; foreign currency only on Federation imports (phase 2).
net_amount, tax_amount, gross_amount	numeric(14,2)	Frozen at issuance.
reference_document_id	uuid FK document (nullable)	Primary upstream document; corrections use document_link.
content_hash	char(64)	SHA-256 over the canonical header and lines; verified on sync; hash chain in phase 2.
origin, device_seq	enum, bigint	ONLINE OFFLINE; the device sequence for gap checks.
engine_version, snapshot_version	text, bigint	For till receipts: what the till computed with (ADR-31, doc 32).

Table document_line

Lines are immutable with their header; facts needed to reproduce a price or stock effect are stored, not looked up later.

Column	Type	Meaning and rules
document_line_id, document_id, line_no		Frozen order.
sku_id, batch_id	uuid FK	Batch mandatory on stock-effecting lines; the price-resolution batch on a receipt.
uom_code, qty	FK, numeric(14,3)	As transacted.
unit_price, mrp_applied, control_price_applied, cap_reason	numeric, enum	NONE MRP_LOWEST MRP_BARCODE MRP_PICKED CONTROL_PRICE.
discount_rule_id, discount_amount, tax_rate_percent, tax_amount, line_total		Frozen; the rule id is the sole source of any reduction (ADR-14).
unit_cost_at_issue	numeric(14,4)	Stock-effecting lines; entity average at issue (carried in the receipt bundle for reporting).
loss_category	enum (nullable)	WOF lines: the F-01 categories.
reference_line_id	uuid FK (nullable)	Upstream line.
negative_ack, picked, weight_kg	boolean, boolean, numeric	Receipt-line facts from the till (E-09; picker; weighed goods).

Table document_link

Typed relationships between documents: the compensating-correction pattern.

Column	Type	Meaning and rules
from_document_id, to_document_id	uuid FK	The later document points at the original.

Column	Type	Meaning and rules
link_type	enum	REVERSES (full void or reversal: at most once per original, only while reversible per the type's state machine); CREDITS (credit note or refund against an invoice or receipt; carries amount); DEBITS; ADJUSTS (count adjustment against a count); SETTLES (payment receipt against specific invoices, with amount; never oldest-first for entities – I-07; customer receipts allocate oldest-first through M7's allocation table instead, ADR-46); DISPUTES (discrepancy or claim against a GRN or DN); SUPERSEDES (drafts only).
amount	numeric(14,2) (nullable)	For SETTLES, CREDITS, DEBITS.
created_at, created_by		

- The net position of any original is the original plus the sum of its linked corrections; reports never re-read a corrected original alone.

Table document_state_history

Append-only record of every transition; the current status is the last row.

Column	Type	Meaning and rules
history_id, document_id		
from_status, to_status	text	
occurred_at, occurred_local, received_at	timestampz	P9.
actor_user_id, device_id	uuid FK	
reason_code, reason_text	text	Mandatory for void, reject, reverse and any transition the type's design marks.

Table document_attachment

Photographs and files for discrepancy, claim and write-off documents (D-08, F-05).

Column	Type	Meaning and rules
attachment_id, document_id		
object_key, content_hash	text, char(64)	Blob in object storage; uploaded via pre-signed PUT, possibly after the document issues.
status	enum	PENDING COMPLETE FAILED; a decision that depends on evidence waits for COMPLETE.
captured_at, captured_by, device_id		

5.5 Issuance protocol and series scope

Step	Action	Guarantee
1	Validate the draft (parties, lines, ceilings, permissions incl. offline_allowed when offline).	No invalid document takes a number.
2	Take the next number: online, a single-row UPDATE ... RETURNING on numbering_series in the issuance transaction; offline, the holder device increments its persisted counter in the same local transaction, fsync before acknowledgement.	Gapless and monotonic; a crash leaves both or neither.
3	Freeze totals, compute content_hash, set issued_at, write the DRAFT→ISSUED state row and the first audit event.	One transaction.
4	Apply stock movements for stock-effecting types.	Movements cite the document.
5	On sync, central verifies the hash and device_seq continuity and advances its view; a gap is an anomaly, never silently absorbed; central never re-numbers.	The device's number is final.

Scope	Document types	Counter holder
TILL_POSITION	RCT (incl. void and same-session refund), CPR when recorded at a till	The device assigned to the position

Scope	Document types	Counter holder
LOCATION	GRN (shop), CNT, WOF, RPK (shop), XFR receipt	The shop's primary till (ADR-47)
ENTITY	ORD (buyer), DN, INV, CN, DN2, PRC (seller), DISC, CLM (issuer), GRN (warehouse), ADJ, RPK (warehouse), XFR issue, OPB, CPR (office)	Central; online only

Invariants for Part C.

- C-I1. No column of document or document_line changes after issued_at is non-null, except status through document_state_history; a database trigger enforces it and lines have no UPDATE privilege. Module extension tables inherit the trigger (I-F10).
- C-I2. (series_id, doc_number) is unique and dense; verified nightly and on sync.
- C-I3. A document's series scope matches its type; a GRN captured at a shop draws from the receiving shop's LOCATION series; an order from the buyer's ENTITY series.
- C-I4. A bilateral document has a non-null counterparty and a relationship in force on business_date.
- C-I5. Every issued document has exactly one DRAFT→ISSUED history row with occurred_at = issued_at.
- C-I6. A correction never changes the original; it is discoverable only through document_link.
- C-I7. content_hash recomputed from stored rows equals the stored value; verified on sync and by a sampled nightly job.
- C-I8. A customer payment receipt (CPR) is an M7 document; trading payment receipts (PRC) hold entity payers only (I-F7).

6. Part D – Append-only audit log

Table audit_event_type

The event catalogue as data (doc 07 §3 plus the events the module designs added).

Column	Type	Meaning and rules
event_type_code	varchar(40) PK	e.g. VOID, REFUND, WRITE_OFF_REQUEST, WRITE_OFF_WITNESSED, WRITE_OFF_APPROVE, STOCK_ADJUST, NEGATIVE_STOCK_SALE, DRAWER_OPEN_NO_SALE, ROLE_CHANGE, USER_DEACTIVATE, SERIES_REGISTER, SERIES_HOLDER_CHANGE, DEVICE_POSITION_CHANGE, SYNC_ANOMALY, REGULATOR_GRANT, CREDIT_LIMIT_CHANGE, CONTROL_PRICE_ENTER, CONTROL_PRICE_CAP_APPLIED, MRP_PICKER_USED, MODE_SWITCH_SHOP_MGMT, PHONE_CHANGE (customer), ACCOUNT_LIMIT_BREACH, CHEQUE_BOUNCED, DSAR_FULFILLED, API_CLIENT_REGISTERED.
severity	enum	INFO REVIEW ALERT; REVIEW and ALERT feed exception queues.
reviewer_role_template	FK role (nullable)	The named reviewer (N-06); required before a REVIEW/ALERT type can be raised in production.
offline_capturable	boolean	All till events are.

Table audit_event

One row per audited action; insert-only; partitioned monthly; retained indefinitely (J-06, 24B N-6).

Column	Type	Meaning and rules
audit_id	uuid PK	UUIDv7 at the source device.
event_type_code	FK	
occurred_at, occurred_local, received_at	timestampz	Partition key received_at.
owner_entity_id, location_id	uuid FK	RLS applies to audit reads as to operational reads.
actor_user_id, actor_system	uuid, varchar(40)	One of the two.
device_id, till_position_id	uuid FK (nullable)	
subject_table, subject_id, document_id	varchar(40), uuid, uuid	Polymorphic by design.
before_state, after_state	jsonb	Minimal diff; never credential material or personal-data values.
reason_code, reason_text, witness_user_id	text, uuid	Mandatory where the type demands; witness for write-offs (remote witness for single-staff shops).
device_seq, correlation_id	bigint, uuid	Gap detection; grouping of one user action.
prev_hash, row_hash	char(64) (nullable)	Reserved for the phase-2 hash chain.

- The application role holds INSERT only; no trigger, function or job may UPDATE or DELETE; partitions are detached to cold storage, never dropped.
- Personal data: the audit log stores identifiers, not personal-data values. A customer phone change records the customer id and the fact of change; the values live in the customer table under PDPA rules (ADR-27).
- Relationship to the sync event stream: the till's outbox carries issued documents with their lines and state rows, stock movements and audit events; all are append-only at both ends and replayed idempotently by primary key. The audit log is complete at central only after sync, which is why federation-level exception reports carry the per-entity freshness stamp (O-03).

Invariants for Part D.

- D-I1. No UPDATE or DELETE ever succeeds on audit_event from the application role; a deployment test asserts the privilege set.
- D-I2. Every document state transition has an audit event with the same correlation_id.
- D-I3. Every REVIEW or ALERT type has a reviewer role template before it can be raised in production.
- D-I4. (device_id, device_seq) is unique and dense per device; gaps raise SYNC_ANOMALY.
- D-I5. Audit reads obey the RLS classes; the Federation and auditors read view-only.

7. Part E – Event backbone and sync state (kernel)

Table event_outbox

Transactional outbox and archive (ADR-30): the system of record for domain events; monthly partitions; indefinite retention.

Column	Type	Meaning and rules
event_id	uuid PK	UUIDv7.
event_type	text	Dotted, versioned, e.g. grn.confirmed.v1.
occurred_at, occurred_local, received_at	timestampz	Partition key received_at.
source, source_seq	text, bigint	central or a device id; dense per source.
owner_entity_id, location_id	uuid FK	Routing and RLS.
aggregate_type, aggregate_id	text, uuid	
correlation_id, causation_id, actor_user_id, engine_version		
payload	jsonb	Small; identifiers and changed facts; never personal-data values.
published_at	timestampz (nullable)	Set by the relay; the relay role holds UPDATE on this column only.

Table event_inbox

Per-consumer idempotency record.

Column	Type	Meaning and rules
consumer, event_id	PK	
applied_at, outcome, last_error		APPLIED SKIPPED FAILED.

Table device_sync_cursor

The only server-side sync state per device (doc 32); row-locked during ingestion.

Column	Type	Meaning and rules
device_id	uuid PK	
last_applied_seq, last_batch_id, last_ack_at		Highest sequence durably applied; idempotent replies.
snapshot_version_reported, app_version_reported		
pending_instructions	jsonb	Revoke, floor notice, re-request-from-seq, series holder change.

Table sync_quarantine

Events that could not be applied: stored raw, never lost, never blocking.

Column	Type	Meaning and rules
quarantine_id	uuid PK	
device_id, device_seq, event_id, raw_payload		
reason_code, reason_text	text	Schema, hash, duplicate id, sequence anomaly.
received_at, resolved_at, resolved_by		Resolution is an audited manual action; never deleted.

Table snapshot_change_log

Per-location change log producing snapshot deltas; retained for the delta window.

Column	Type	Meaning and rules
location_id, version	PK	Dense per location.
table_name, row_id, op	text, uuid, enum	UPSERT TOMBSTONE.
apply_from, urgent	date, boolean	Business-date activation; urgent for control prices, rule withdrawals, account suspensions, permission revocations.
created_at	timestampz	

Table idempotency_key

Result store for mutating web and API commands.

Column	Type	Meaning and rules
idempotency_key, user_id	PK, uuid	Scoped by principal.
request_hash, response_status, response_body, created_at		A reused key with a different body is rejected; rows expire after 24 hours.

Invariants for Part E.

- E-I1. Every business transaction that changes a table in parts A–C or a module schema inserts at least one event_outbox row in the same transaction.
- E-I2. (source, source_seq) is unique and dense per source; gaps not explained by sync_quarantine raise SYNC_ANOMALY.
- E-I3. device_sync_cursor.last_applied_seq never decreases except through the audited sequence-reset procedure.
- E-I4. A location's snapshot_change_log versions are dense; a delta is the contiguous range from the device's version + 1.

8. Part F – Module schema register and cross-module invariants

One PostgreSQL schema per module (ADR-29). Parts A–E are authoritative for kernel and shared tables; for every other table the implementation guide named here holds the authoritative DDL. No module writes another module's schema (R4).

Schema	Owner	Tables	Authoritative DDL
kernel	Kernel	document_type, numbering_series, document, document_line, document_link, document_state_history, document_attachment, audit_event, audit_event_type, event_outbox, event_inbox, device_sync_cursor, sync_quarantine, snapshot_change_log, idempotency_key, notification_log, config_item, config_value, scheduled_job, job_run, shedlock	Parts C–E; 19A
party	M1	entity, entity_relationship, location, till_position, device	Part A; 21A §3.1
security	M1	app_user, permission, permission_catalogue_version, role, role_permission, user_role, sod_pair, external_grant	Part A; 21A §3.2
catalogue	M2	uom, supplier, sku, sku_uom_conversion, sku_barcode, sku_alias, sku_image, tag, sku_tag, tax_category, tax_rate, batch, location_assortment, duplicate_suspect	Part B; 22A §3
pricing	M3	price_list, price_list_line, discount_rule, control_price, mrp_policy, adjustment_type, label_job	Part B; 23A §3
trading	M4	doc_order, doc_order_line, allocation_run, doc_delivery, doc_delivery_drop, doc_delivery_line, doc_grn, doc_grn_line, doc_discrepancy, doc_claim, doc_invoice, doc_payment_receipt (entity payers only), cheque, relationship_exposure, posting_map, transfer_request; views v_invoice_party, v_invoice_line_party	24A §3 as amended by CR-27-2
inventory	M5	stock_lot, stock_movement, entity_sku_cost, repack_recipe, loss_category, variance_threshold, loss_tolerance, expiry_window, count_task, expiry_task, pick_list; extensions doc_count_line, doc_adjustment, doc_writeoff, doc_repack, doc_transfer, doc_opening_balance	Part B (lot, movement, recipe); 25A §3
pos	M6 (central mirror)	doc_receipt, doc_receipt_tender, till_session, cash_movement, device_health; the till's local schema mirrors parts C–E plus cache tables	26A §10 (central), §3 (till)
customers	M7	customer, customer_phone, customer_consent, customer_tag, customer_account, account_posting, allocation, doc_customer_payment, data_subject_request, banking_record, banked_session	27A §3
reporting	M8	projection_state, location_freshness, receipt_fact, receipt_line_fact, sales_daily, cash_daily, stock_position, stock_movement_fact, loss_fact, trade_document_fact, dispute_register, exposure_snapshot, customer_receivables, till_session_fact, pricing_review_fact, fleet_health, exception_item, report_definition, report_run, export_job, dashboard_tile, user_dashboard	28A §3 (rebuildable projections; never a source of truth)
integration	M9	api_scope, api_client, api_credential_history, api_call_log, posting_map_registry, journal_export, journal_line, notification_template, notification_rule, provider_config, e_invoice_submission, webhook_subscription	29A §3
hello	Template	greeting	17A §12 (removed before pilot)

8.1 Document type ownership

Owner	Document types
M4 Trading Documents	ORD, DN, GRN, DISC, CLM, INV, CN, DN2, PRC
M5 Inventory	CNT, ADJ, WOF, RPK, XFR, OPB
M6 POS	RCT (incl. void and refund receipts)
M7 Customers	CPR

8.2 Cross-module invariants

- I-F1. A location's `location_assortment` equals the union of SKUs on its owner's published retail list, SKUs with a lot at the location, the owner's LOCAL SKUs and SKUs tagged core-range; the nightly reconcile reports drift.
- I-F2. `batch.printed_mrp` is non-null when the SKU's `has_printed_mrp` is true and the receipt line quantity is positive; null otherwise is legitimate.
- I-F3. `stock_lot` is unique per (location, batch, condition); a DAMAGED lot carries the unit cost of its GOOD sibling.
- I-F4. `stock_movement.movement_seq` is dense per (location, source).
- I-F5. Every receipt line's `batch_id` names the lot the sale deducts; a sale may take a lot negative; availability never counts negative lots (ADR-44, ADR-45).
- I-F6. No table in the reporting schema is read by a transactional decision; every reporting table is rebuildable from `event_outbox`.
- I-F7. A customer payment receipt (CPR) is an M7 document; `trading.doc_payment_receipt` holds entity payers only.
- I-F8. Control prices, segregation pairs and posting maps are data with their own tables, never configuration-register items (P10 of doc 17).
- I-F9. An API client's scopes never exceed the granting administrator's permissions; the same containment rule as roles.
- I-F10. Every module-owned document extension table inherits the kernel immutability trigger and is partitioned by the same key as the base.
- I-F11. Only the primary till issues LOCATION-series documents; other tills capture drafts (ADR-47).
- I-F12. Customer repayments allocate oldest-first through `customers.allocation`; entity payments settle specific invoices through `SETTLES` links (I-07, ADR-46).

9. Cross-cutting conventions

Concern	Convention
Naming	snake_case; singular table names; _id for UUID keys, _code for human codes; every table has created_at.
Keys	UUIDv7 everywhere (P2); natural keys are unique constraints, never primary keys.
Language columns	Three columns per text field (P7); ICU collations si_icu, ta_icu, en_icu; trigram indexes for partial match; NFC normalisation at write; sort by the display language; Western Arabic digits in every language. Full-text search analysis (OpenSearch-class) is phase 2 and reads from these columns.
Time	timestampz UTC plus a local reading (P9); business_date from the location's day-close.
Money and quantity	numeric(14,2) LKR; numeric(14,4) unit cost and price; numeric(14,3) quantity; cash rounding is a receipt-level field.
Soft delete	No DELETE (P10); every reference table has a status with a terminal value.
Partitioning	document, document_line, extension tables, stock_movement, account_posting, document_state_history, event_outbox, audit_event, api_call_log, reporting facts: range-partitioned by month on received_at; partitions created three months ahead; detached to cold storage under the tiering policy (doc 14 DR-6), never dropped.
Row-level security	Enabled and forced on every operational table from the first migration with the classes of 3.7; a table without a policy fails the schema test; PARTY reads go through masking views.
Extension tables	Keyed one-to-one on document_id in the owning module's schema; inherit the immutability trigger and partition key.
Till local database	SQLite via Room, encrypted with a keystore-held key; the same logical model for what the till writes (parts C–E plus receipt and session tables) and read-only cache tables for what it receives; same column names and types.
Database roles	app_rw (module code; INSERT-only on ledgers), app_relay (UPDATE published_at only), app_report and app_analyst (SELECT on reporting on the replica), migration role (DDL).

10. Decision requests

#	Point	Status	Decider	If different
DR-1	Financial-year reset of numbering	Closed: no reset (24B N-1)	–	–
DR-2	GRN outside the fiscal gapless obligation	Closed: yes (24B N-2)	–	–
DR-3	Control price scope by tag as well as by SKU; lower applies when both in force	Open (assumed both; lower applies)	Federation catalogue steward; Federation legal (doc 10 L-04)	Tag scope dropped: every gazetted category must be enumerated per SKU by the steward; or a specificity rule replaces lower-applies in M3's CeilingResolver
DR-4	Batch identity global; cost on the lot	Closed: adopted (ADR-38)	–	–
DR-5	Synthetic batches for non-batch-tracked SKUs	Closed: adopted (doc 22)	–	–
DR-6	Alignment with doc 17 module ownership	Closed (v0.2)	–	–
DR-7	Single-staff write-off witness rule	Closed: remote witness with mandatory photos (doc 25 DR-4)	–	–
DR-8	Control-price operational owner	Open (assumed Federation catalogue steward)	Federation (doc 10 H-05)	Without a named owner the ceiling table goes stale and the hard block enforces wrong ceilings
DR-9	Reporting schema excluded from parts A–E invariants (CR-28-1)	Closed: adopted (I-F6)	–	–

11. Decisions recorded from this model (doc 09)

ADR	Decision	Source
ADR-37	Trading documents are one module (M4); document base and numbering are kernel	Doc 17 v0.2.1
ADR-38	Batch identity in M2; stock lot and cost in M5; synthetic batches	Doc 17, doc 22
ADR-39	Stateless application tier	Doc 14 §1.3
ADR-40	Updates through a device-owner management agent; no till-to-till sharing in phase 1	Doc 31, doc 17

ADR	Decision	Source
ADR-41	Multi-MRP resolution AUTO_LOWEST default; control price as a separate date-effective ceiling, hard block	Multi-MRP study, doc 23
ADR-42	Numbering decisions N-1 to N-6 adopted; tax advisor review advisory	24B
ADR-43	In-store labels encode SKU code plus weight or batch, never price	Doc 22 DR-6
ADR-44	Availability = non-negative GOOD lots minus issued-undispatched delivery quantities; allocation does not reserve	Doc 25 DR-1
ADR-45	Oversell accepted: lots may go negative; exception raised; never rejected	Doc 25, doc 26
ADR-46	Customer repayments allocate oldest-first; captured offline at the till	Doc 27 DR-1, DR-2
ADR-47	Only the primary till issues LOCATION-series documents; other tills capture drafts	Doc 26 DR-2
ADR-48	Report catalogue as decision-driven data; projections rebuildable; freshness on every federation figure	Doc 28
ADR-49	Public API exposes module operations under the same scope rules; clients are principals; one write in phase 1	Doc 29
ADR-50	Native Android for the till with a dedicated till track	Doc 30 DR-1, doc 14

12. Traceability to the requirements baseline

Model element	Requirements and decisions
Part A	FR-PTY-010 to 070; FR-SEC-010 to 070; FR-GOV-010 to 060; ADR-02, 03, 11, 18, 26; A, M, N sections; CR-21-1, CR-21-2
Part B: SKU, units, barcodes, images, tags, tax, assortment	FR-SKU-010 to 110; FR-LOC-020; ADR-07; B section; doc 20 Appendix A; CR-22-1, CR-22-2; doc 22 DR-7; ADR-43
Part B: batch, lot, movement	FR-INV-010 to 100; FR-LOSS-010; ADR-08, ADR-38, ADR-44, ADR-45; E-03, E-04, E-09; F-01; CR-25-1
Part B: ceilings, lists, rules, policies	FR-PRC-010 to 130; G-03, G-09, G-10; ADR-14, ADR-31, ADR-41; CAA Act s.20(5); CR-23-1
Part C	FR-BIL-010 to 150; doc 03; ADR-11; 24B; I-07; D-05, D-07, D-08; F-03 to F-05; H-02/H-03; CR-24-1, CR-27-1, CR-27-2
Part D	FR-SEC-050; N-05, N-06, N-07, N-09; doc 07 §3; J-06; ADR-27
Part E	ADR-30; doc 19 §6; doc 32; doc 14 §1.3
Part F	ADR-29; doc 17 rule R4; CR-28-1, CR-29-1; ADR-46, ADR-47

13. Abbreviations

Term	Meaning
AI	GS1 Application Identifier: the numbered data fields inside a GS1 barcode (01 product number, 10 batch, 17 expiry).
CAA	Consumer Affairs Authority of Sri Lanka; gazettes maximum retail prices.
CPR / PRC	Customer payment receipt (M7) / entity payment receipt (M4).
FEFO	First expired, first out: the picking rule for expiry-tracked goods.
GRN	Goods received note: the receiving entity's record of what arrived; the moment ownership transfers.
GS1	The global standards body for product identification and barcodes.
ICU	International Components for Unicode: the library providing correct sorting and comparison for Sinhala and Tamil.
IRD	Inland Revenue Department of Sri Lanka.
LKR	Sri Lankan rupee.
MFA / TOTP	Multi-factor authentication / time-based one-time password.
MPCS	Multi-purpose cooperative society: the legal entity that owns retail shops.
MRP	Maximum retail price printed on a pack; a batch attribute.
PDPA / PII	Personal Data Protection Act No. 9 of 2022 / personally identifiable information.
POS	Point of sale: the till.
RBAC	Role-based access control.

Term	Meaning
RLS	Row-level security: database policies that filter rows by the caller's scope.
SKU	Stock keeping unit: the smallest distinctly tracked and counted item.
UOM	Unit of measure.
UTC	Coordinated Universal Time.
UUIDv7	Universally unique identifier, version 7: time-ordered, safe to generate on disconnected devices.
VAT	Value Added Tax.

End of document. v0.3.1 is the data model of record; 18B is withdrawn. The next revision follows the Stage 4 documents.